One-Variable Inequalities: Catch the Sign Flip
Answer Key

Grades 6–7 Mathematics

Quick Answers

1. $(7, \infty)$
2. $[-6, \infty)$
3. (a) the solution = $(-\frac{9}{2}, \infty)$, (b) least integer solution = -4
4. $(4, \infty)$
5. $[4, \infty)$
6. (b) correct solution = $(-4, \infty)$, —
7. (b) correct solution = $(-\infty, -4)$, boundary value shown in the key = -4, —
8. (a) solution = $(7, \infty)$, —

What is verified

5 of 8 problems fully machine-checked against SymPy.
— marks an answer only you can judge — problems 6, 7, 8. The worked solution states what a correct response must contain.

Curriculum

Level: Grades 6–7 Mathematics
7.EE.B.4 – *problems 1, 2, 3, 4, 5, 6, 7, 8*
Difficulty 1–4 of 5 across 13 tagged checks.

Common wrong answers

3. *If they got* -5*: did not reverse the inequality, so picked an integer from x < -4.5 instead of x > -4.5*
7. *If they got* -7*: added 3 to the right side instead of subtracting, giving -2x > 14*

1. Solve the inequality: $6x > 42$
Solution: Divide both sides by 6 (a *positive* number, so the symbol stays the same): $x > \frac{42}{6}$, so the solution is $\boxed{x > 7}$

2. Solve the inequality: $-5x \leq 30$
Solution: Divide both sides by -5. Dividing by a *negative* number reverses the symbol, so $\leq$ becomes $\geq$: $x \geq \frac{30}{-5}$, giving $\boxed{x \geq -6}$

3. Solve the inequality $-4x < 18$; then give the least integer solution.
Solution (a): Divide both sides by -4 and reverse the symbol: $x > \frac{18}{-4}$, so $\boxed{x > -\frac{9}{2}}$
Solution (b): $-\frac{9}{2} = -4.5$, and the least integer greater than -4.5 is $\boxed{-4}$
Common wrong answer: a student who forgets to reverse the symbol gets $x < -4.5$ and picks -5, which is not a solution: $-4(-5) = 20$ and $20 < 18$ is false.

4. Solve the inequality: $3x - 7 > 5$
Solution: Add 7 to both sides: $3x > 12$. Divide by 3 (positive, no reversal): $\boxed{x > 4}$

5. Solve the inequality: $-2x + 9 \leq 1$
Solution: Subtract 9 from both sides: $-2x \leq -8$. Divide by -2 and reverse the symbol: $x \geq \frac{-8}{-2}$, so $\boxed{x \geq 4}$

6. Priya's work on $-3x < 12$: she wrote $x < -4$.
Solution (a): *(instructor-judged)* Priya divided both sides by -3 but did **not** reverse the inequality symbol. Full credit for any sentence naming the missing reversal when dividing by a negative.
Solution (b): Divide both sides of $-3x < 12$ by -3 and reverse the symbol: $\boxed{x > -4}$

7. Deshaun's work on $-2x + 3 > 11$: he wrote $-2x > 14$, then $x < -7$.
Solution (a): *(instructor-judged)* The error is in step 1: Deshaun *added* 3 to the right side ($11 + 3 = 14$) instead of *subtracting* 3 from both sides. The correct step is $-2x > 11 - 3$, that is $-2x > 8$. (His step 2 reversal was actually correct.)
Solution (b): From $-2x > 8$, divide by -2 and reverse the symbol: $x < \frac{8}{-2}$, so $\boxed{x < -4}$
The boundary value of the correct solution comes from $-2x + 3 = 11$, whose solution is $x = -4$: machine-checked as $\boxed{-4}$

8. Freezer experiment: temperature $14 - 3h$ after h hours; when is it below $-7°$F?
Solution (a): Require $14 - 3h < -7$. Subtract 14: $-3h < -21$. Divide by -3 and reverse the symbol: $h > \frac{-21}{-3}$, so $\boxed{h > 7}$
The temperature is below $-7°$F after more than 7 hours.
Solution (b): *(instructor-judged)* The symbol reversed because both sides were divided by -3, a negative number; dividing by a negative reverses the order of every pair of numbers. Any clear statement of this rule earns full credit.